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eMethods

Recruitment

The biobank was initially established at Amsterdam UMC. Participants are recruited through five main routes: (1) tertiary memory clinics, including Alzheimer Center Amsterdam[1]; (2) memory clinics of regional hospitals; (3) nursing homes specialized in young-onset dementia (YOD), self-referral, and existing research studies, including YOD-COHORT[2] ('JMD-studie' in Dutch),[3] which is part of the national YOD-INCLUDED consortium addressing a broad range of YOD-specific research questions. Eligible individuals are approached and by their treating physician, a research physician or research assistant. Eligible individuals who express interest are invited for a single research baseline visit.

Informed consent procedure

For the DemenTREE Biobank, written informed consent for participation is mandatory. With consent to DemenTREE, residual biomaterial from related studies may also be stored in the DemenTREE Biobank for future research. Furthermore, participants consent to data linkage with medical files of memory clinics or nursing homes, with Statistics Netherlands (Centraal Bureau voor de Statistiek; CBS[4]), and with the Personal Records Database (Nederlandse Basisregistratie Personen; BRP[5]), the latter two containing demographic and socioeconomic data. In case a participant is affected by dementia, a relative or caregiver is asked to co-sign the consent together with the participant.

In addition, participants consent to participation in Alzheimer Register Nederland (formerly ABOARD Cohort),[6] a national register study focusing on the trajectory of Alzheimer disease and dementia. After registration, participants receive an email to confirm their enrollment, which becomes active upon confirmation. Participation enables longitudinal follow-up through annual online questionnaires, although questionnaire completion is optional and not required for participation in DemenTREE. In addition, a medical case report form is sent to the participant's treating physician to collect clinical information from the medical record, including retrospectively available biomarker and imaging data. This centralized data collection is designed to reduce duplicate data entry by treating physicians and enable reuse of harmonized clinical data across participating research studies.

Beyond the standard elements of informed consent, participants are also invited to provide additional consent on the following topics:

- Use of photographs of hands and face (for research purposes only);
- Data linkage to other Dutch (medical) registration systems, such as health insurers for data on healthcare use;
- Additional blood draw for isolation of peripheral blood mononuclear cells (PBMCs), for example to generate induced pluripotent stem cells (iPSCs);
- Permission to be contacted for future research purposes;
- Permission to be contacted for participation in clinical trials.

The offering of these additional consent options differs by recruitment site. For example, the collection of photographs and the blood draw for isolation of PBMCs is currently limited to participants recruited through Amsterdam UMC.

Data collection and storage

Demographics, health and family information

Participants provide demographic information (age, country of birth, education, housing) and health information (medical history, cognitive symptoms, diagnosis). Education is classified using the Dutch Verhage scale[7] and converted to low, medium and high.[8] For participants who visited a memory clinic or live in a nursing home, additional clinical data are obtained from their medical records, including clinical presentation, neurological examination, diagnosis, and, if performed in the clinical setting, results on genetic testing. At baseline, cognitive status is assessed with the Montreal Cognitive Assessment (MoCA).[9] If this is not feasible, MoCA or Mini-Mental State Examination (MMSE)[10] scores within six months prior to baseline are extracted from medical records.

Family history is recorded for dementia subtypes, Parkinson's disease, and other neurodegenerative diseases (e.g., multiple sclerosis, and amyotrophic lateral sclerosis), as well as psychiatric and cardiovascular diseases. Pedigrees are constructed with documentation of sex, age (or age at death), and, where applicable, diagnoses, and corresponding age at onset (*Figure 2*). Pedigrees are drafted manually and subsequently converted to a digital, coded format for analysis in R.

Diagnostic classification
For analyses, dementia subtypes were grouped into 5 categories based on the most common clinical diagnoses: (1) AD, including typical AD, corticobasal syndrome due to AD (CBS-AD), logopenic variant primary progressive aphasia (lvPPA), and posterior cortical atrophy (PCA); (2) frontotemporal dementia, including behavioral (bvFTD) and right temporal variant FTD (rtvFTD); (3) Lewy body disease, including dementia with Lewy bodies (DLB) and Parkinson's dementia (PD); (4) semantic variant of primary progressive aphasia (svPPA); and (5) other diagnoses, including vascular dementia (VaD), nonfluent variant PPA (nfvPPA), progressive supranuclear palsy (PSP), corticobasal degeneration (CBD), and other rare neurodegenerative diseases.[11–17]

Photographs of hands and face
Among participants recruited through Amsterdam UMC, photographs of the palmar and dorsal side of the hands and frontal and profile views of the face are collected and securely stored to document phenotypic features potentially associated with genetic etiologies. These images are not shared externally or published without additional explicit participant consent.

Biomaterial collection, processing and storage
Biomaterial collection consists of a single blood draw for DNA, plasma, and serum. Buccal swabs are used as an alternative DNA source if blood sampling is not feasible or is undesirable. Cerebrospinal fluid (CSF) is not collected as part of the study protocol. However, when a lumbar puncture is performed as part of clinical care or related research studies, surplus CSF can be used for research and/or an additional 10 ml CSF may be drawn for research purposes. In participants from the Amsterdam UMC site, we collect Mononuclear Cell Preparation Tubes (CPT's) for PBMC isolation following standard protocols.[18] PBMCs are stored at -80 °C for a maximum of three days, after which they are transferred to liquid nitrogen storage at the Amsterdam UMC Biobank core facility.

DNA is isolated from whole blood or buccal cells using the FlexSTAR+ system and subsequently stored at 4°C until analysis. Plasma, serum, and CSF samples are processed locally following standardized published pre-analytics.[19,20] Biomaterial is centrifuged within two hours at 1800 g for 10 minutes at room temperature and aliquoted into 0.5 ml portions into 2D-barcoded Micronic polypropylene cryovials. All samples are stored at the Amsterdam UMC Biobank core facility at -70 to -80 degrees Celsius pending analysis.

Familial aggregation analysis
Standardized multigenerational pedigrees were reconstructed from reported family history data, including both enrolled participants and non-enrolled relatives, and visualized using in-house R scripts. Relationships and reported disease status were used to classify relatives as first-degree (parents and full siblings) or second-degree (grandparents and parental siblings). For the case-control analyses, one affected proband was included per family and compared with unrelated partner controls. Family history was coded as the presence or absence of at least one affected relative within each degree of relatedness. Subtype-specific analyses were performed for AD, FTD, DLB, and VaD. For AD analyses, relatives reported to have AD or dementia of unspecified type were combined. Logistic regression models estimated the odds of being an affected proband according to family history, with results expressed as ORs with 95% CIs. Models were unadjusted for other covariates because of the limited sample size. Models were not estimated when cell counts were insufficient.

Biomarker analysis
The case-only analysis was restricted affected participants by grouping clinical diagnoses into three different categories (AD-spectrum: AD, CBS-AD, lvPPA, PCA; FTLD-spectrum: bvFTD, CBD, nfvPPA, PSP, rtvFTD, svPPA; and other dementias: DLB, PD, VaD, cerebral amyloid angiopathy), and comparing plasma biomarker levels between these groups using linear regression models, adjusting for age and sex as covariates.

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eTables

eTable 1. Enrolled participants per study site.

	Participants, N. (%)			
Inclusion site	**Inclusion route**	**Total (n=830)**	**Affected (n=387)**	**Unaffected (n=443)**
Amsterdam UMC	Academic hospital	583 (70.2)	251 (64.9)	332 (74.9)
Spaarne Gasthuis	Regional hospital	106 (12.8)	51 (13.2)	55 (12.4)
Dijklander Ziekenhuis	Regional hospital	33 (4)	19 (4.9)	14 (3.2)
Jeroen Bosch Ziekenhuis	Regional hospital	24 (2.9)	24 (6.2)	0 (0)
Lisidunahof, Beweging 3.0	Nursing home	13 (1.6)	6 (1.6)	7 (1.6)
De Wever	Nursing home	6 (0.7)	2 (0.5)	4 (0.9)
YOD-COHORT	Research study	65 (7.8)	34 (8.8)	31 (7.0)

eTable 2. Availability of materials and data.

	Participants, N. (%)		
	Total (n=830)	**Affected (n=387)**	**Unaffected (n=443)**
Available biomaterials			
Blood	664 (80.1)	318 (82.4)	346 (78.1)
Buccal swab	153 (18.5)	63 (16.4)	90 (20.4)
Cereobrospinal fluid	17 (4.1)	17 (4.4)	0 (0)
PBMCs	209 (57.3)	204 (62)	5 (13.9)
National registry (ABOARD Cohort)			
Accepted registry invitation	343 (45.3)	257 (72.4)	86 (21.4)
Filled in online questionnaire	139 (18.4)	93 (26.2)	46 (11.4)

Abbreviation: PBMCs = peripheral blood mononuclear cells.

eTable 3. Detailed diagnoses of affected patients.

	Participants, N. (%)
	Affected (n=387)
Alzheimer's disease	**274 (70.8)**
Alzheimer's disease	257 (93.8)
Corticobasal syndrome due to Alzheimer's disease	2 (0.7)
Posterior cortical atrophy	5 (1.8)
Logopenic variant of primary progressive aphasia	10 (3.6)
Frontotemporal dementia	**27 (7)**
Behavioral variant frontotemporal dementia	20 (74.1)
Frontotemporal dementia-amyotrophic lateral sclerosis	1 (3.7)
Frontotemporal dementia unspecified	2 (7.4)
Right temporal variant frontotemporal dementia	4 (14.8)
Lewy Bodies Disease	**24 (6.2)**
Dementia with Lewy Bodies	22 (91.7)
Parkinson's disease dementia	2 (8.3)
Semantic variant of primary progressive aphasia	**20 (5.2)**
Semantic variant of primary progressive aphasia	20 (100)
Other	**35 (9)**
Cerebral amyloid angiopathy	2 (5.7)
Corticobasal degeneration	3 (8.6)
Homozygotic *XPA* variant	1 (2.9)
Huntington's disease	1 (2.9)
Non-fluent variant of primary progressive aphasia	8 (22.9)
Limbic-predominant age-related TDP-43 encephalopathy	3 (8.6)
Primary progressive aphasia unspecified	1 (2.9)
Progressive supranuclear palsy	4 (11.4)
Vascular dementia	12 (34.3)
Unknown	**7 (1.8)**
Unknown	7 (100)

eTable 4. Demographic and disease characteristics of proxy-related reports in family history data.

			First-degree relatives		Second-degree relatives		
	Total	**Probands & partners**	**Siblings**	**Parents**	**Parental siblings**	**Grandparents**	**Other**
Total	11234 (100)	783 (7)	1749 (15.6)	1218 (10.8)	3815 (34.0)	2004 (17.8)	1665 (14.8)
Female	5023 (48.6)	395 (50.4)	706 (47.3)	609 (50.0)	1601 (47.5)	1002 (50.0)	710 (48.3)
Died	5597 (63.5)	8 (1.2)	271 (22.0)	979 (83.0)	2121 (86.7)	1964 (99.7)	254 (19.2)
Age, *mean ±SD*	57.7 ±20.8	65 ±7.8	66.3 ±8.8	83.8 ±7	80.8 ±8.7	87 ±5.3	33.4 ±14.5
Age at death, *mean ±SD*	71.9 ±19.5	66.8 ±7.9	53 ±26	77.1 ±12.2	70.1 ±22.6	75.7 ±12.6	65.4 ±27.2
Reported disease status							
Cardiovascular accident (CVA)[a]	139 (1.3)	6 (0.8)	32 (2.0)	69 (5.8)	21 (0.6)	9 (0.5)	2 (0.1)
Neurodegenerative disease	1404 (13.6)	376 (49)	146 (9.0)	293 (24.5)	389 (11.4)	141 (8)	59 (3.7)
Amyotrophic lateral sclerosis	9 (0.1)	0 (0)	3 (0.2)	2 (0.2)	1 (0)	1 (0.1)	2 (0.1)
Dementia	1255 (12.1)	336 (43.8)	123 (7.6)	263 (22.0)	362 (10.6)	128 (7.3)	43 (2.7)
Alzheimer's disease	521 (5.0)	270 (35.2)	49 (3.0)	79 (6.6)	87 (2.5)	25 (1.4)	11 (0.7)
Dementia with Lewy bodies	41 (0.4)	23 (3.0)	5 (0.3)	8 (0.7)	4 (0.1)	0 (0)	1 (0.1)
Dementia unknown	594 (5.7)	0 (0)	54 (3.3)	148 (12.4)	259 (7.6)	103 (5.9)	30 (1.9)
Frontotemporal dementia	53 (0.5)	32 (4.2)	10 (0.6)	7 (0.6)	4 (0.1)	0 (0)	0 (0)
Other type of dementia	45 (0.4)	34 (4.4)	4 (0.2)	2 (0.2)	5 (0.1)	0 (0)	0 (0)
Vascular dementia	55 (0.5)	12 (1.6)	9 (0.6)	25 (2.1)	8 (0.2)	0 (0)	1 (0.1)
Mild cognitive impairment	12 (0.1)	7 (0.9)	2 (0.1)	2 (0.2)	1 (0)	0 (0)	0 (0)
Multiple sclerosis	19 (0.2)	0 (0)	5 (0.3)	1 (0.1)	4 (0.1)	1 (0.1)	8 (0.5)
Other neurodegenerative disease	11 (0.1)	10 (1.3)	1 (0.1)	0 (0)	0 (0)	0 (0)	0 (0)
Parkinson's disease	95 (0.9)	2 (0.3)	17 (1.1)	32 (2.7)	24 (0.7)	13 (0.7)	7 (0.4)
Psychiatric disorder	286 (2.8)	21 (2.7)	54 (3.3)	53 (4.4)	54 (1.6)	16 (0.9)	88 (5.6)

Table shows N (%) unless otherwise specified. Probands (n=403) and partners (n=380) include one patient or unrelated partner per family history, respectively (*Fig. 2*). Proxy reports are attributed to either the corresponding proband or partner. Other types of dementia include PPA variants, CAA, and Korsakov dementia, among others. Other neurodegenerative diseases include progressive supranuclear palsy, corticobasal syndrome, and Huntington's disease. [a]CVA includes CVA of unknown origin, ischemic CVA, and haemorrhagic CVA. Psychiatric disorders include psychiatry of unknown origin, depression, bipolar disorder, anxiety disorder, attention-deficit/hyperactivity disorder, autism spectrum disorder, psychosis, and/or schizophrenia. Available N: died = 8,817; age = 2,950, age at death = 1,944, reported disease status = 10,347. *Abbreviations: SD = standard deviation.*

eTable 5. Familial aggregation of dementia across proband dementia subtypes.

Disease in proband	Predictor	N case	N control	OR [95% CI]	p-value
Alzheimer's disease	First-degree relatives	269	1972	2.39 [1.82-3.14]	<.001
	Second-degree relatives	364	3702	1.54 [1.24-1.92]	<.001
Frontotemporal dementia	First-degree relatives	7	2234	6.42 [1.23-33.41]	.03
	Second-degree relatives	3	4063	7.22 [0.65-79.93]	.11
Dementia with Lewy bodies	First-degree relatives	9	2232	0 [0-Inf]	.98
	Second-degree relatives	3	4063	0 [0-Inf]	.97
Vascular dementia	First-degree relatives	27	2214	6.02 [1.75-20.74]	.004
	Second-degree relatives	5	4061	0 [0-Inf]	.99

Abbreviations: CI = confidence interval; OR = odds ratio.

eTable 6. Familial aggregation of Alzheimer's disease in first- and second-degree relatives in relation to proband disease status: unadjusted and covariate-adjusted logistic regression models.

Model	Predictor in relatives	β	95% CI	p-value	n
First-degree relatives					
Unadjusted	Alzheimer's disease	0.871	0.602 – 1.147	<.001	2241
Age-adjusted	Alzheimer's disease	0.865	0.575 – 1.163	<.001	1808
	Age	-0.002	-0.008 – 0.004	.52	
Sex-adjusted	Alzheimer's disease	0.906	0.634 – 1.185	<.001	2102
	Sex	0.089	-0.085 – 0.263	.32	
Second-degree relatives					
Unadjusted	Alzheimer's disease	0.434	0.214 – 0.656	<.001	4066
Age-adjusted	Alzheimer's disease	0.287	-0.124 – 0.697	.17	688
	Age	0.009	2.90×10^{-4} – 0.018	.05	
Sex-adjusted	Alzheimer's disease	0.418	0.196 – 0.644	<.001	3840
	Sex	0.009	-0.118 – 0.137	.89	

β values represent logistic regression coefficients on the log-odds scale. Age was modelled per 1-year increase. Sex was included as a categorical variable, with female as the reference category; the coefficient for sex (i.e., male sex) represents the difference in log odds relative to females. Sample sizes vary across models because analyses were restricted to participants with complete data for all variables included in each model. *Abbreviations: CI = confidence interval.*

eTable 7. Characteristics of DemenTREE participants included and not included in the biomarker analyses.

Participants, No. (%)			
Characteristic	**Biomarker N=254**	**No biomarker N=576**	**P-value**
Female	131 (51.6)	304 (52.8)	.81
Age inclusion, mean ±SD	61.1 ±12.9	61.3 ±13.6	.92
Education			.20
High	141 (57.8)	331 (59.3)	
Middle	69 (28.3)	129 (23.1)	
Low	34 (13.9)	98 (17.6)	
Country of birth: Netherlands	237 (94.4)	516 (91.2)	.56
Family history of dementia	134 (71.7)	280 (67.6)	.37
Family type specified			.002
Patients	102 (40.3)	285 (49.7)	
Partners	84 (33.2)	155 (27)	
Children	38 (15)	70 (12.2)	
Siblings	20 (7.9)	19 (3.3)	
Parents	5 (2)	11 (1.9)	
Other	4 (1.6)	34 (5.9)	
Cognitive syndrome			.08
Cognitively normal	145 (57.1)	282 (49)	
Subjective cognitive decline	7 (2.8)	9 (1.6)	
Mild cognitive impairment	17 (6.7)	46 (8)	
Dementia	85 (33.5)	239 (41.5)	
Diagnosis			.64
Alzheimer disease	70 (68.6)	204 (73.4)	
Frontotemporal dementia	7 (6.9)	20 (7.2)	
Lewy body disease	8 (7.8)	16 (5.8)	
Semantic variant PPA	8 (7.8)	12 (4.3)	
Other diagnosis	9 (8.8)	26 (9.4)	
Clinical characteristics			
MoCA at baseline, mean ±SD	23.2 ±6.2	22.7 ±5.9	.36
Study site			<.001
Tertiary hospital	215 (84.6)	362 (62.8)	
Regional hospital	30 (11.8)	133 (23.1)	
Nursing home	7 (2.8)	12 (2.1)	
YOD-COHORT	1 (0.4)	64 (11.1)	
Self-referral	1 (0.4)	5 (0.9)	

Percentages and summary statistics were calculated using available data.Differences between participants included and not included in the biomarker analyses were assessed using χ^2 tests for categorical variables and analysis of variance for continuous variables. *Abbreviations: PPA = primary progressive aphasia, MoCA = Montreal Cognitive Assessment; SD = Standard Deviation; YOD-COHORT = Young Onset Dementia Cohort.*

eTable 8. Demographics and plasma biomarker levels of affected and unaffected participants.

	Median [IQR]		
	Affected participants (n=102)	**Unaffected relatives (n=152)**	**P-value**
Age inclusion, *mean ±SD*	65.5 (7.4)	58.2 (14.8)	<.001
Female, *n (%)*	49 (48.0%)	82 (53.9%)	.43
Plasma Aβ42 (pg mL^{-1})	5.80 (4.92–6.65)	6.40 (5.17–7.51)	.007
Plasma Aβ40 (pg mL^{-1})	113.1 (93.1–125.2)	119.5 (101.6–133.2)	.02
Plasma Aβ42/40	0.05 (0.05–0.06)	0.05 (0.05–0.06)	.15
Plasma GFAP (pg mL^{-1})	127.70 (90.40–166.70)	62.96 (44.55– 89.58)	<.001
Plasma NfL (pg mL^{-1})	19.20 (14.12–28.31)	10.29 (6.34–13.96)	<.001
Plasma pTau217 (pg mL^{-1})	0.10 (0.05–0.17)	0.03 (0.02–0.04)	<.001

The affected group included participants with Alzheimer disease (n=70) and non-Alzheimer's disease dementias (n=32), including Lewy body disease (n=8), frontotemporal dementia (n=7), semantic variant of primary progressive aphasia (n=8), and other causes (n=9). The unaffected relatives include 68 family members and 84 partners. *Abbreviations: Aβ = amyloid-beta, GFAP = glial fibrillary acidic protein, pTau217 = tau phosphorylated at threonine 217, NfL = neurofilament light chain, IQR = interquartile range.*

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eTable 9. Comparisons of plasma biomarker levels between affected and unaffected at the median age of affected participants (66 years).

Biomarker	Effect type	Estimate	Lower 95% CI	Upper 95% CI	P value
Aβ42/40	Difference	−0.001	−0.004	0.003	.70
GFAP	Ratio	1.72	1.54	1.92	<.001
NfL	Ratio	1.75	1.54	1.99	<.001
pTau217	Ratio	2.41	2.05	2.85	<.001

Comparisons of plasma biomarker levels between affected and unaffected participants at age 66 years, the median age of affected participants. Estimates were derived from linear regression models including age, participant group, and their interaction. For GFAP, NfL, and pTau217, models were fit on the log-transformed scale; estimates are ratios of mean biomarker levels between groups, back-transformed to the original scale. For Aβ42/40, models were fit on the original scale; the estimate is the absolute difference in mean Aβ42/40 between groups. *Abbreviations: Aβ42/40 = amyloid-β42/β40; GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; pTau217 = phosphorylated tau at threonine 217.*

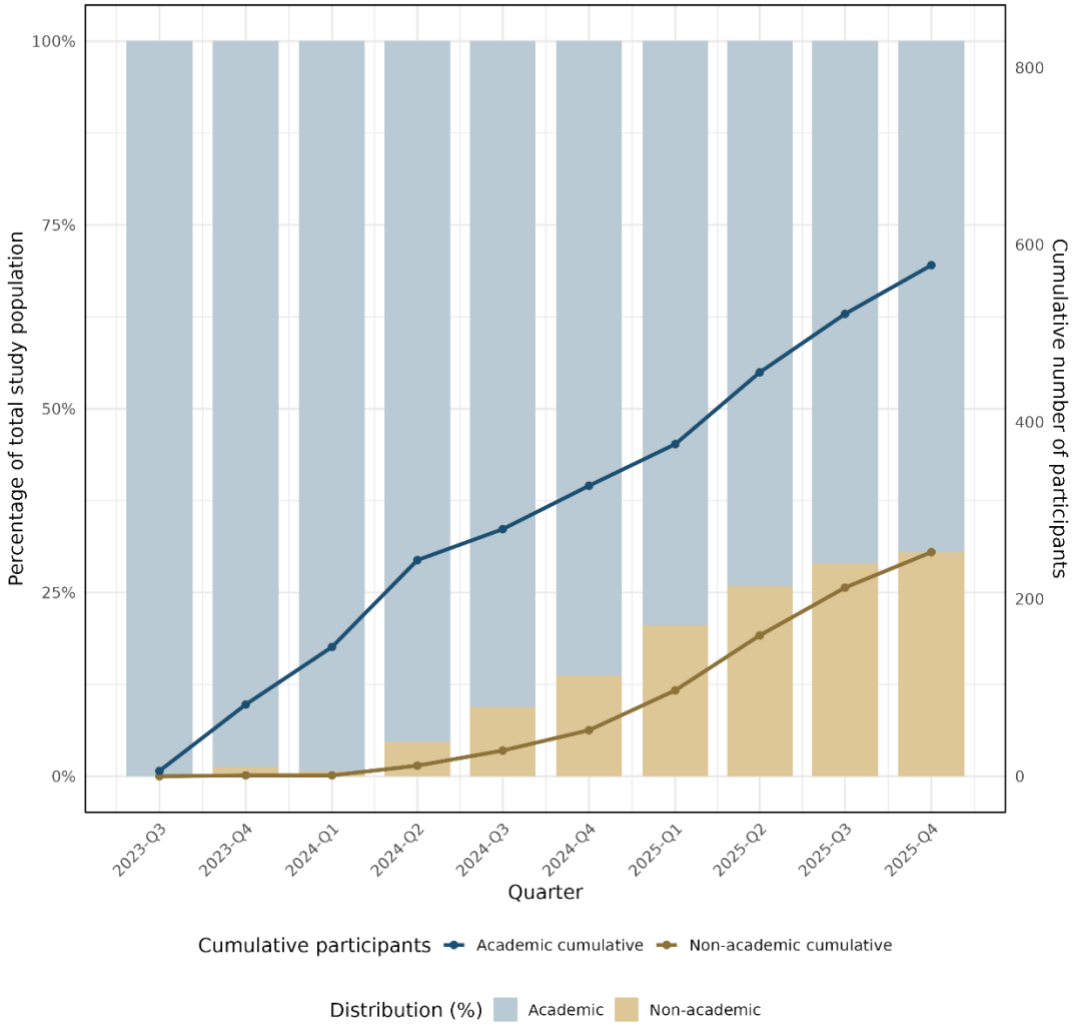
eTable 10. Comparisons of plasma biomarker levels across diagnostic groups among affected participants.

Comparison	Biomarker	Effect type	Estimate	Lower 95% CI	Upper 95% CI	P value
AD vs FTLD	Aβ42/40	Difference	−0.008	−0.017	0.001	.07
	GFAP	Ratio	1.09	0.85	1.41	.67
	NfL	Ratio	0.50	0.37	0.68	<.001
	pTau217	Ratio	2.38	1.54	3.67	<.001
AD vs Other	Aβ42/40	Difference	−0.005	−0.016	0.006	.53
	GFAP	Ratio	1.03	0.75	1.41	.98
	NfL	Ratio	0.57	0.38	0.84	.003
	pTau217	Ratio	2.04	1.19	3.49	.006
FTLD vs Other	Aβ42/40	Difference	0.003	−0.010	0.016	.83
	GFAP	Ratio	0.94	0.65	1.36	.91
	NfL	Ratio	1.14	0.73	1.80	.77
	pTau217	Ratio	0.86	0.46	1.60	.83

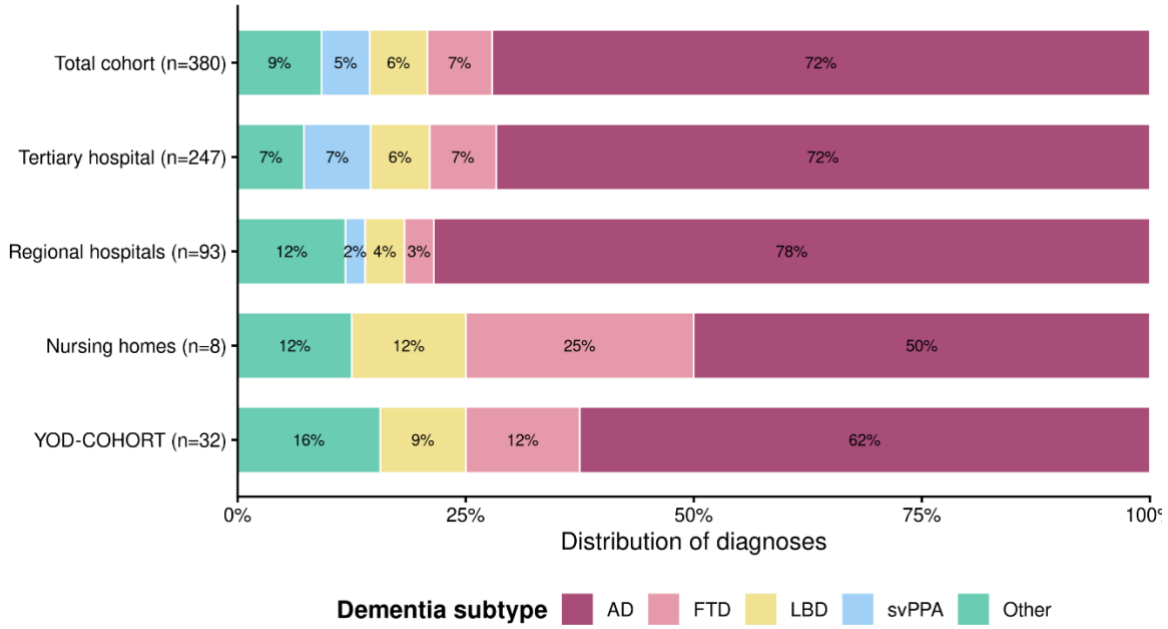
Pairwise comparisons of plasma biomarker levels between diagnostic subgroups of affected participants (AD-spectrum, FTLD-spectrum, and other dementias), adjusted for age and sex. For GFAP, NfL, and pTau217, models were fit on the log-transformed scale; estimates are ratios of mean biomarker levels between groups, back-transformed to the original scale. Ratios can be converted to percentage differences as (ratio − 1) × 100%. For Aβ42/40, models were fit on the original scale; estimates are absolute differences in mean Aβ42/40 between groups. P values and 95% CIs were adjusted for multiple comparisons using the Tukey method. *Abbreviations: Aβ42/40 = amyloid- β42/β40; GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; pTau217 = phosphorylated tau at threonine 217. AD = Alzheimer disease; FTLD = frontotemporal lobar degeneration.*

eFigures

eFigure 1. Distribution of participants from academic versus non-academic hospitals.

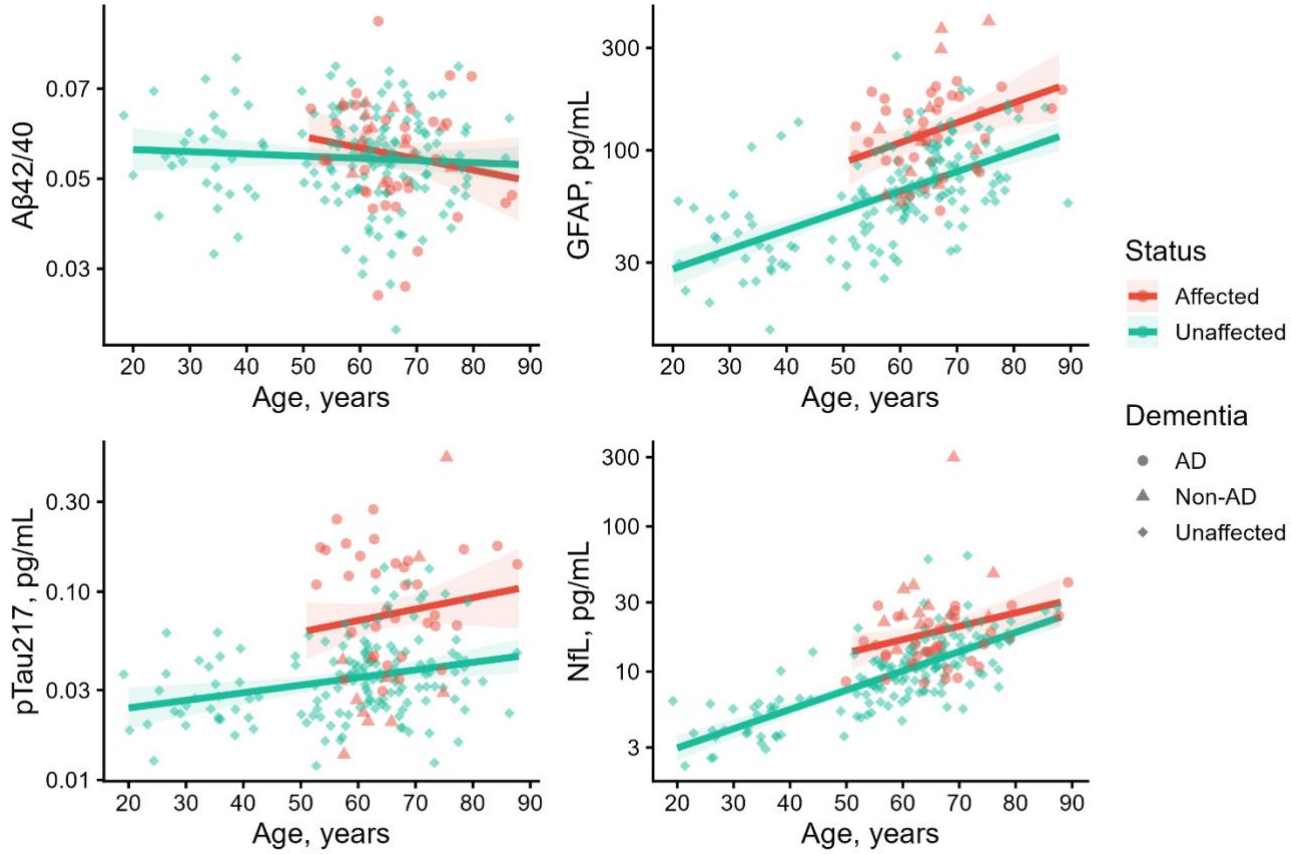


eFigure 2. Distribution of dementia subtypes stratified by inclusion route.



Self-referred participants (n=3) enrolled at Amsterdam UMC were included in the tertiary hospital group. *Abbreviations: AD = Alzheimer disease; FTD = frontotemporal dementia, LBD = Lewy body disease; svPPA = semantic variant of primary progressive aphasia; YOD = young-onset dementia.*

eFigure 3. Plasma biomarker levels in affected participants and unaffected relatives.



Predicted plasma biomarker concentrations and 95% confidence intervals are plotted for each marker from the fitted models. Predictions are restricted to the observed age range within each group. Predicted values were back-transformed to the original scale for visualization. Axes are displayed on a log10 scale to appropriately represent the distribution of biomarker concentrations. Ages are jittered by ±2 years for anonymization. *Abbreviations: Aβ = amyloid-beta, AD = Alzheimer disease; GFAP = glial fibrillary acidic protein; NfL = neurofilament light chain; pg/mL = picogram/mililiter; pTau217 = tau phosphorylated at threonine 217.*